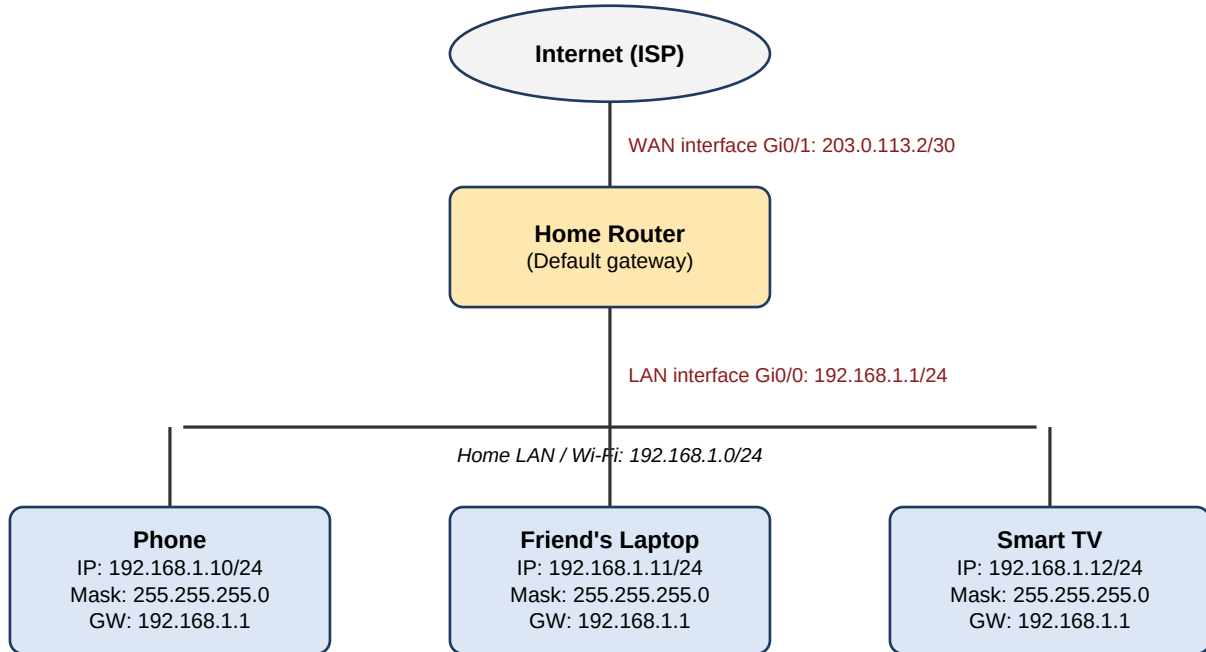


IP Routing and Routing Protocols

Assignment Answers

Question 1: Simple Network Diagram

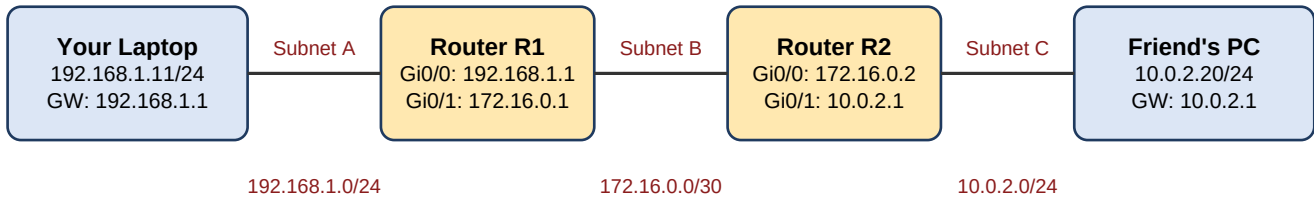
Draw a network with three devices (phone, friend's laptop, smart TV) connected through a router, labelling IP addresses of each device and the router's interfaces.



Explanation: All three devices are in the same private subnet 192.168.1.0/24. The router's LAN interface (192.168.1.1) is their default gateway, so any traffic for the internet is sent to it. The router's WAN interface (203.0.113.2/30, a made-up public address) connects to the ISP. Devices talk to each other directly inside the LAN; the router is used only to reach other networks.

Question 2: Manual IP Routing Simulation

Explain step by step how a packet travels from your laptop to a friend's computer in a different subnet, showing routers passed and routing decisions (made-up IPs).



Routing tables (made-up)

Router	Type	Destination	Next Hop	Interface
R1	C (connected)	192.168.1.0/24	-	Gi0/0
R1	C (connected)	172.16.0.0/30	-	Gi0/1
R1	S (static)	10.0.2.0/24	172.16.0.2	Gi0/1
R2	C (connected)	172.16.0.0/30	-	Gi0/0
R2	C (connected)	10.0.2.0/24	-	Gi0/1
R2	S (static)	192.168.1.0/24	172.16.0.1	Gi0/0

Packet journey: 192.168.1.11 to 10.0.2.20

Step 1 - Laptop decides the destination is remote. The laptop ANDs the destination 10.0.2.20 with its own mask 255.255.255.0 and gets 10.0.2.0, which is not its own network (192.168.1.0). So it sends the packet to its default gateway, 192.168.1.1. It uses ARP to find the gateway's MAC address and builds a frame: **Src IP 192.168.1.11, Dst IP 10.0.2.20**, Dst MAC = R1's Gi0/0 MAC.

Step 2 - Router R1 (first hop). R1 receives the frame on Gi0/0, removes the Layer-2 header and looks up 10.0.2.20 in its routing table. The checked entries are: 192.168.1.0/24 (no match), 172.16.0.0/30 (no match), **10.0.2.0/24 via 172.16.0.2 (match)**. R1 decrements TTL (64 to 63), ARPs for 172.16.0.2 and forwards the packet out Gi0/1 in a new frame.

Step 3 - Router R2 (second hop). R2 receives the packet on Gi0/0 and checks its table. 10.0.2.20 matches **10.0.2.0/24, directly connected via Gi0/1**, so no further next hop is needed. TTL becomes 62. R2 ARPs for 10.0.2.20 and delivers the frame directly to the friend's PC.

Step 4 - Friend's PC receives the packet. The destination IP matches its own address, so the packet is accepted and passed up to the transport layer.

Step 5 - Reply path. The PC sends the reply to its gateway 10.0.2.1 (R2). R2 matches **192.168.1.0/24 via 172.16.0.1**, forwards to R1, and R1 finds 192.168.1.0/24 directly connected on Gi0/0 and delivers it to the laptop.

Key point: The source and destination IP addresses stay the same end to end, while the MAC addresses and TTL change at every hop. Each router makes an independent decision using the longest-prefix match in its own routing

table.

Question 3: Routing Table for a Zomato-style Delivery Network

Create a routing table for a fictional delivery network with three branches in different cities, each on a different subnet.

Design: A central hub router (HQ-R, Gurugram) connects three branch routers over point-to-point /30 links. Each branch has its own LAN for staff, kitchen tablets and rider devices. All addresses are fictional.

Branch	LAN subnet	Branch router WAN IP	HQ link subnet
Delhi	10.1.1.0/24	172.20.1.2	172.20.1.0/30
Mumbai	10.1.2.0/24	172.20.2.2	172.20.2.0/30
Bengaluru	10.1.3.0/24	172.20.3.2	172.20.3.0/30

Routing table on HQ hub router (HQ-R)

Network Address	Subnet Mask	Next Hop	Interface	Type
10.1.1.0	255.255.255.0	172.20.1.2	Gi0/0	Static - Delhi
10.1.2.0	255.255.255.0	172.20.2.2	Gi0/1	Static - Mumbai
10.1.3.0	255.255.255.0	172.20.3.2	Gi0/2	Static - Bengaluru
172.20.1.0	255.255.255.252	Directly connected	Gi0/0	Connected
172.20.2.0	255.255.255.252	Directly connected	Gi0/1	Connected
172.20.3.0	255.255.255.252	Directly connected	Gi0/2	Connected
0.0.0.0	0.0.0.0	203.0.113.1 (ISP)	Gi0/3	Default route

Routing table on Delhi branch router (example)

Network Address	Subnet Mask	Next Hop	Interface
10.1.1.0	255.255.255.0	Directly connected	Gi0/1 (LAN)
172.20.1.0	255.255.255.252	Directly connected	Gi0/0 (WAN)
10.1.2.0	255.255.255.0	172.20.1.1	Gi0/0
10.1.3.0	255.255.255.0	172.20.1.1	Gi0/0
0.0.0.0	0.0.0.0	172.20.1.1	Gi0/0

How it works: A packet from a Delhi rider's device (10.1.1.x) to a Mumbai kitchen tablet (10.1.2.x) goes to the Delhi router, which sends it via the default/specific route to the HQ hub (172.20.1.1). The hub matches 10.1.2.0/24 and forwards it to 172.20.2.2, the Mumbai router, which delivers it locally. Branch routers use the hub as their next hop for everything that is not local (a 'hub-and-spoke' design).

Question 4: Static vs Dynamic Routing

Compare static and dynamic routing (RIP/OSPF) with 3 pros and 3 cons each, using real-world examples.

Static Routing

Pros	Cons
1. No protocol overhead. Routes are typed in by hand, so no CPU, memory or bandwidth is used for updates. Example: your home Wi-Fi router has one default route to the ISP for Netflix, YouTube and WhatsApp traffic.	1. No automatic failover. If a link goes down, traffic keeps going to the dead path until an admin fixes it. Example: a small shop's UPI payment terminal fails if its single static route's link is down.
2. Predictable and secure. The path never changes and there are no route advertisements an attacker could spoof. Example: a bank branch link fixed to its data centre.	2. Does not scale. With dozens of routers, every route on every router must be added manually; one new subnet means many edits.
3. Simple to set up in small networks, and easy to troubleshoot. Example: a small cafe or office with one router and one ISP.	3. Prone to human error. A typo in a next hop can break connectivity, and nothing detects or corrects it.

Dynamic Routing (RIP / OSPF)

Pros	Cons
1. Automatic adaptation and failover. Routers exchange updates and recalculate paths. Example: an ISP backbone (like those serving Jio or Airtel users) keeps your video call alive when one link fails.	1. Overhead. Updates use CPU, memory and bandwidth (RIP sends its full table every 30 seconds; OSPF floods link-state updates).
2. Scales to large networks. New subnets are learned automatically. Example: a food-delivery company with many cities and data centres can add a new region without editing every router.	2. More complex to configure and troubleshoot (areas, metrics, neighbour states in OSPF).
3. Load balancing and best-path selection. OSPF picks the lowest-cost path and can split traffic over equal-cost paths. Analogy: just as a delivery app re-routes a rider around a traffic jam, dynamic routing re-routes packets around congestion or failure.	3. Security and stability risks. Without authentication, a rogue router can inject false routes; RIP is slow to converge and limited to 15 hops.

Conclusion: Use static routing for small, stable networks and simple default routes; use dynamic routing (OSPF inside an organisation) for large or changing networks that need high availability.

Question 5: OSPF Configuration Between Two Routers

Generate a sample OSPF configuration for two routers and explain each line in your own words.

Topology: R1 (LAN 192.168.1.0/24) is connected to R2 (LAN 192.168.2.0/24) through the link 10.0.12.0/30. R1 uses 10.0.12.1 and R2 uses 10.0.12.2 on the link. All routers are in OSPF Area 0.

Router R1 configuration (AI-generated sample, annotated)

Configuration line	What it does
hostname R1	Sets the router's name to R1 so it is easy to identify.
interface GigabitEthernet0/0	Enters configuration mode for the interface that faces R1's LAN.
ip address 192.168.1.1 255.255.255.0	Gives the LAN interface its IP and mask; this is the default gateway for LAN devices.
no shutdown	Turns the interface on (router interfaces are off by default).
interface GigabitEthernet0/1	Enters the interface that connects to R2.
ip address 10.0.12.1 255.255.255.252	Sets the point-to-point link address; /30 allows only 2 usable hosts.
no shutdown	Enables the link interface.
router ospf 1	Starts OSPF process number 1. The number is only locally significant and need not match on the other router.
router-id 1.1.1.1	Gives R1 a unique identity in OSPF. It is used in neighbour relationships and DR/BDR elections.
network 192.168.1.0 0.0.0.255 area 0	Enables OSPF on interfaces inside 192.168.1.0/24 and advertises this LAN. 0.0.0.255 is a wildcard mask (the opposite of a subnet mask). Area 0 is the backbone area.
network 10.0.12.0 0.0.0.3 area 0	Enables OSPF on the link to R2 so that R1 can form a neighbour relationship and exchange routes.
passive-interface GigabitEthernet0/0	Stops sending OSPF hellos on the LAN (no routers there), but the LAN is still advertised. Improves security and reduces traffic.

Router R2 configuration

Configuration line	What it does
hostname R2	Names the router R2.
interface GigabitEthernet0/0	Interface facing R2's LAN.
ip address 192.168.2.1 255.255.255.0	LAN gateway address for the 192.168.2.0/24 network.
no shutdown	Enables the interface.
interface GigabitEthernet0/1	Interface connected to R1.
ip address 10.0.12.2 255.255.255.252	Other end of the /30 link, in the same subnet as R1's 10.0.12.1.
no shutdown	Enables the interface.
router ospf 1	Starts OSPF on R2.
router-id 2.2.2.2	Unique OSPF ID for R2 (must differ from R1).
network 192.168.2.0 0.0.0.255 area 0	Advertises R2's LAN into OSPF Area 0.
network 10.0.12.0 0.0.0.3 area 0	Enables OSPF on the link to R1. Area must match R1 (Area 0) for the neighbours to form.
passive-interface GigabitEthernet0/0	No OSPF hellos on the LAN side.

Verification commands

Command	Purpose
show ip ospf neighbor	Shows whether R1 and R2 have reached the FULL neighbour state.
show ip route ospf	Lists routes learned via OSPF (marked 'O'); R1 should see 192.168.2.0/24 via 10.0.12.2.
ping 192.168.2.1	Tests end-to-end reachability from R1 to R2's LAN.

Summary: Both routers must be on the same subnet on the link, use the same area, and have unique router IDs. After the neighbour relationship forms, they exchange link-state information and each automatically learns the other's LAN without any static routes.